

CLIP

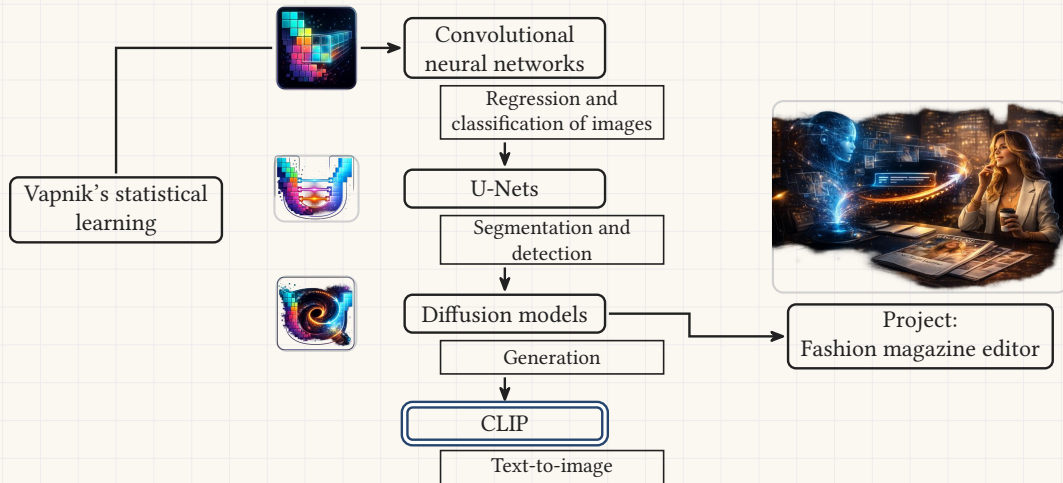
Weekend 3, Saturday morning

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich

Saturday 3 October 2026

Agenda





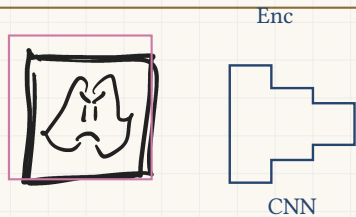
Section 1

CLIP

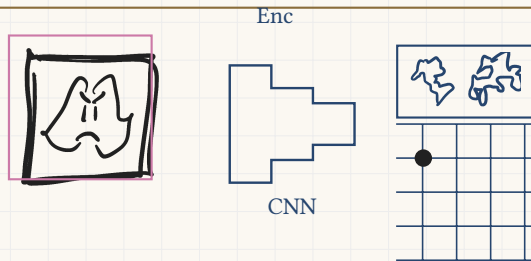
CLIP



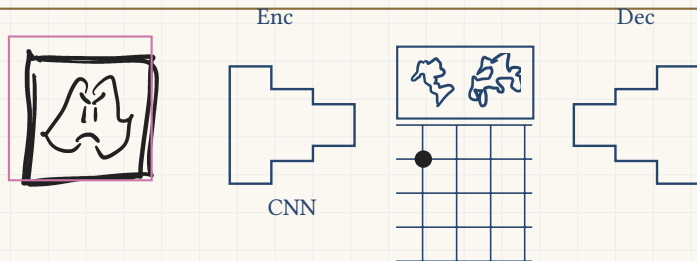
CLIP



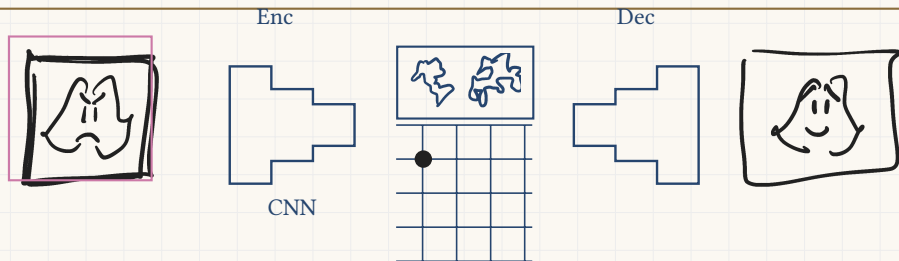
CLIP



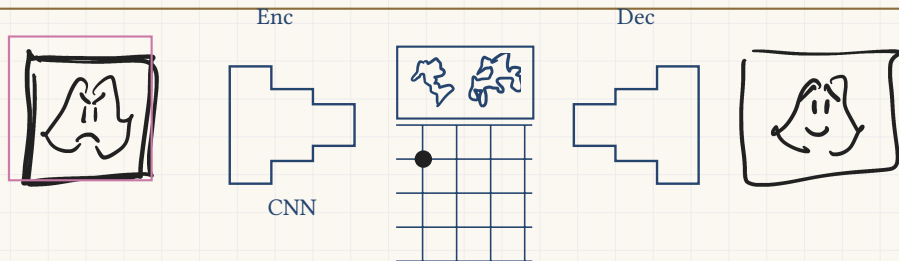
CLIP



CLIP

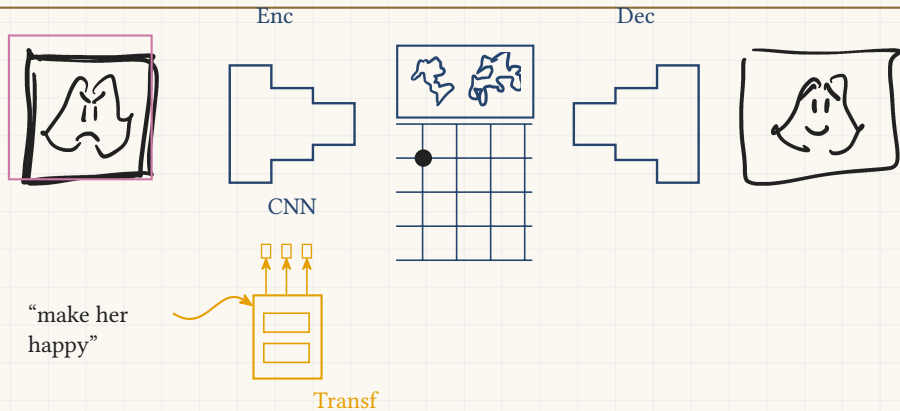


CLIP

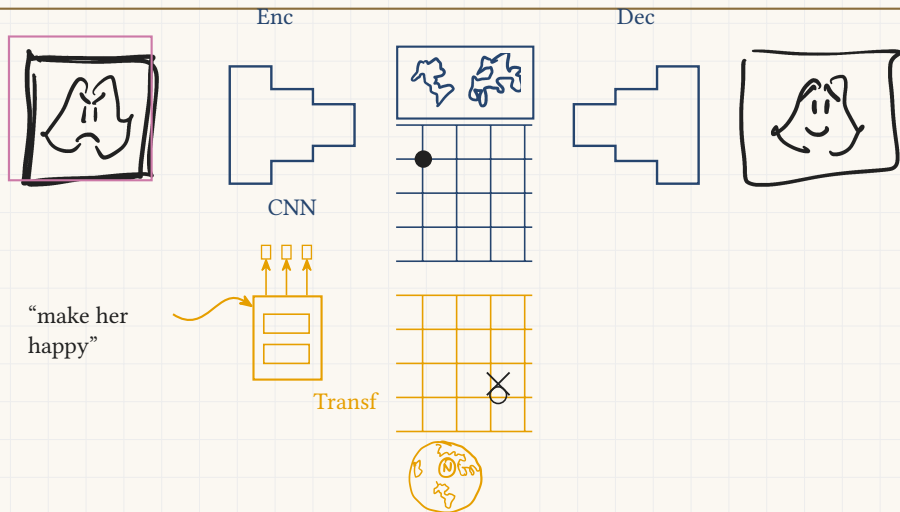


“make her
happy”

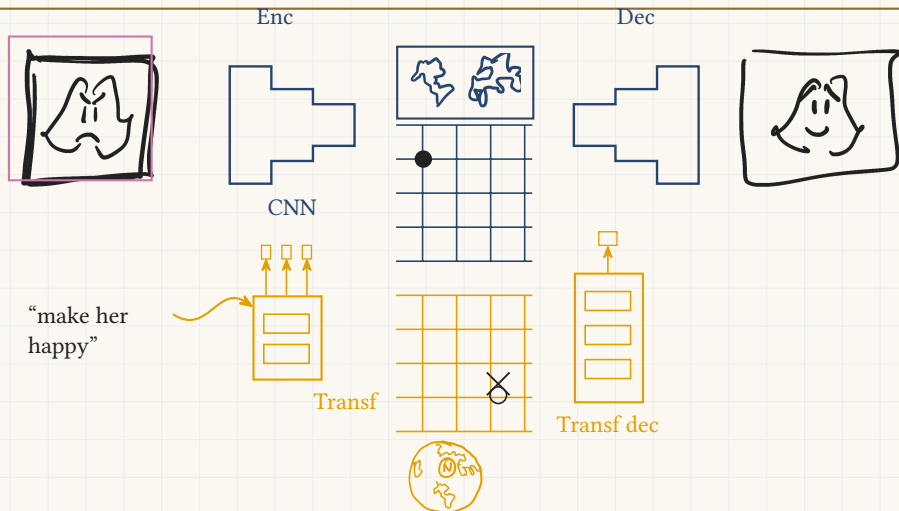
CLIP



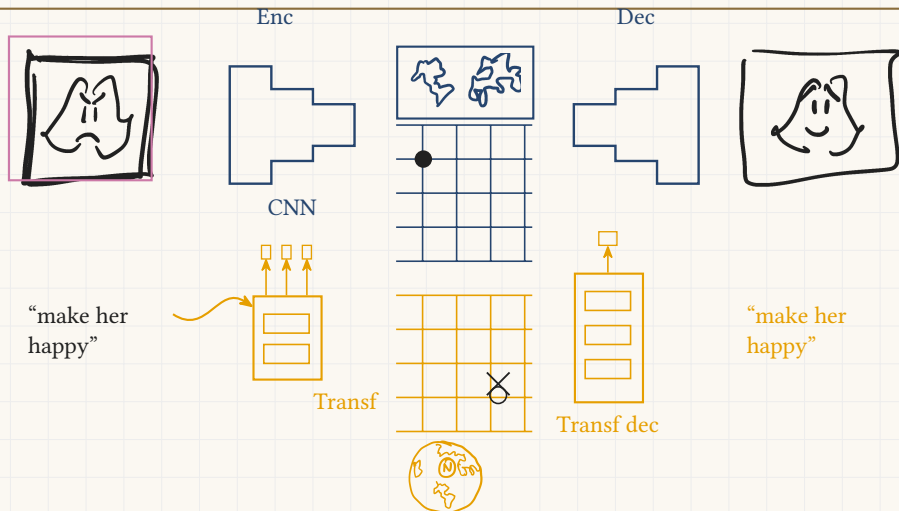
CLIP



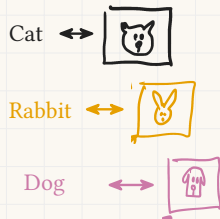
CLIP



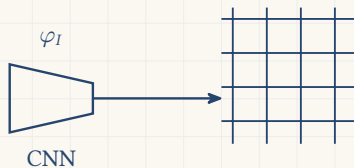
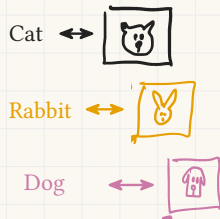
CLIP



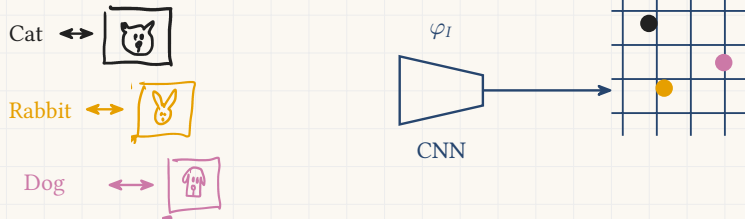
CLIP



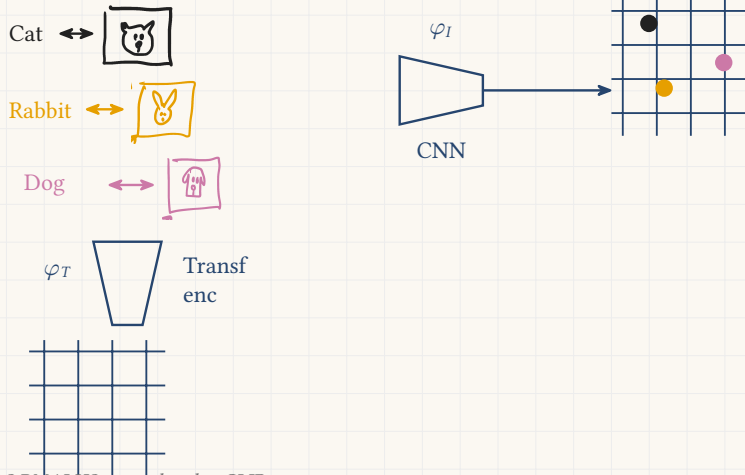
CLIP



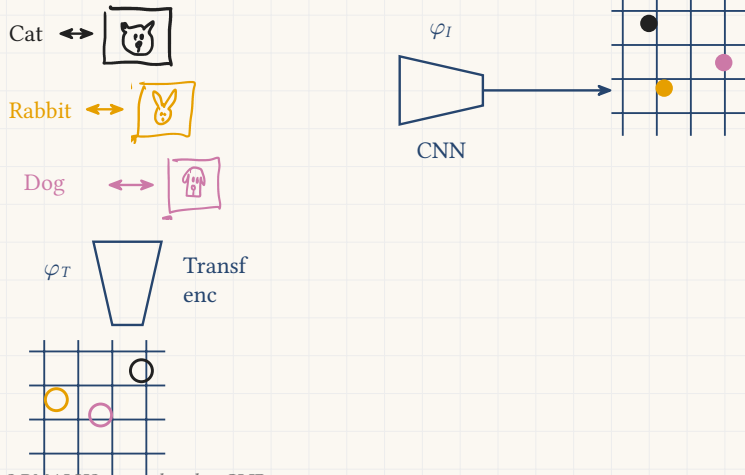
CLIP



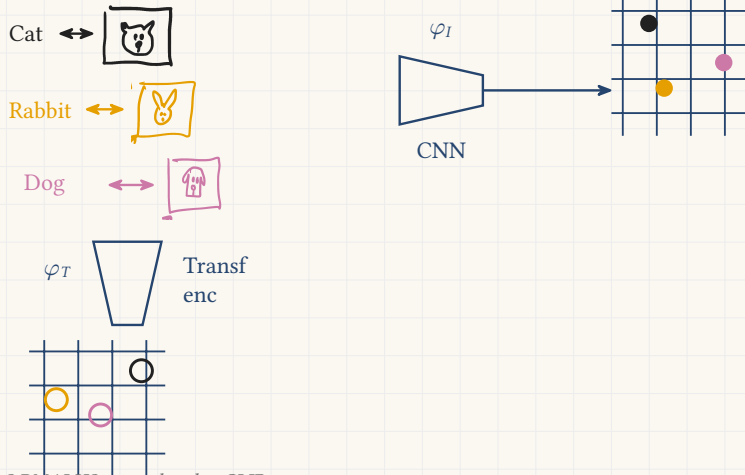
CLIP



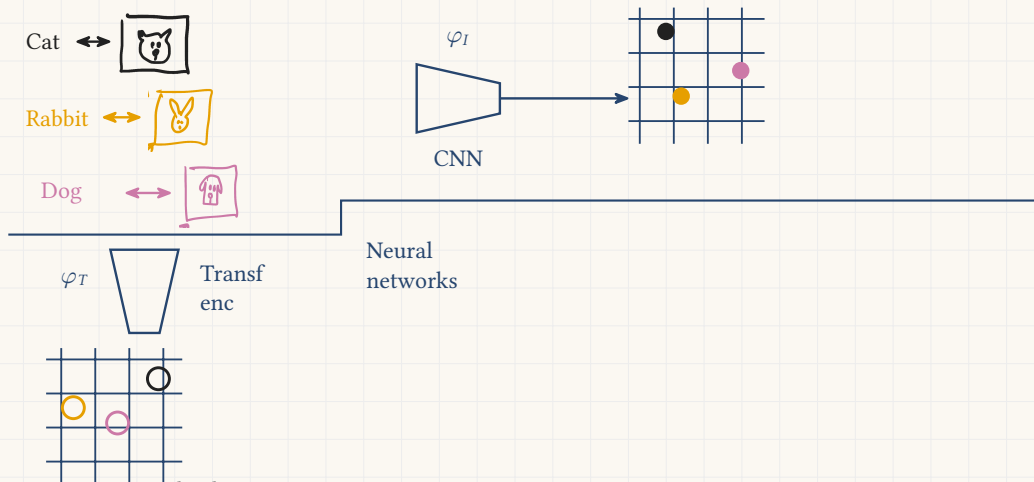
CLIP



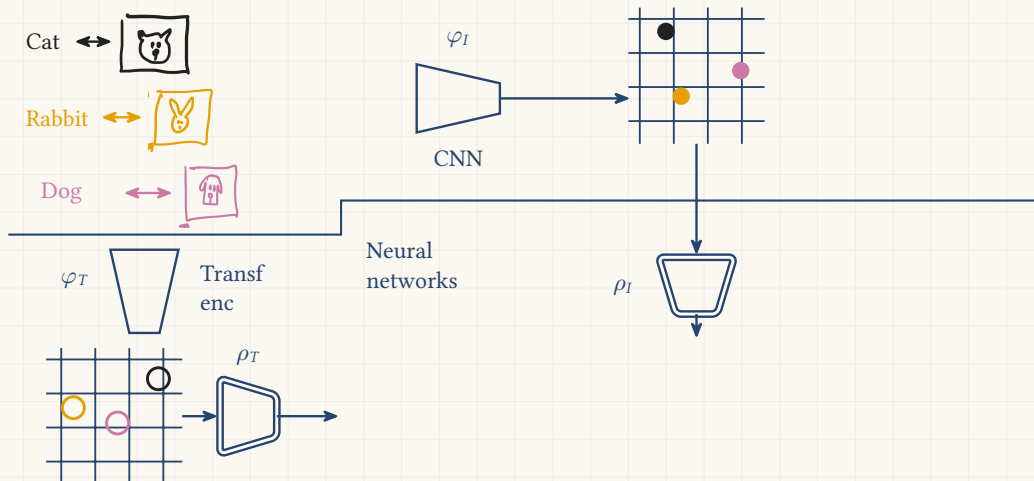
CLIP



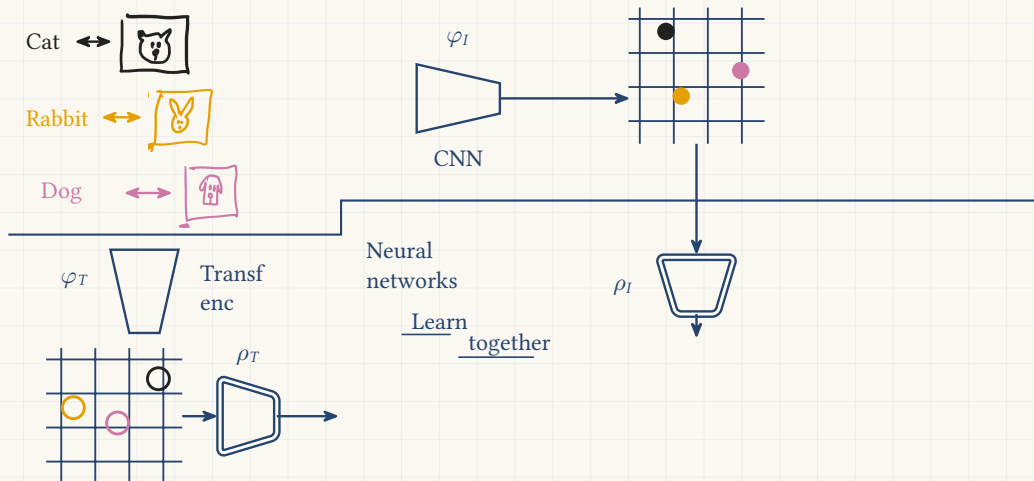
CLIP



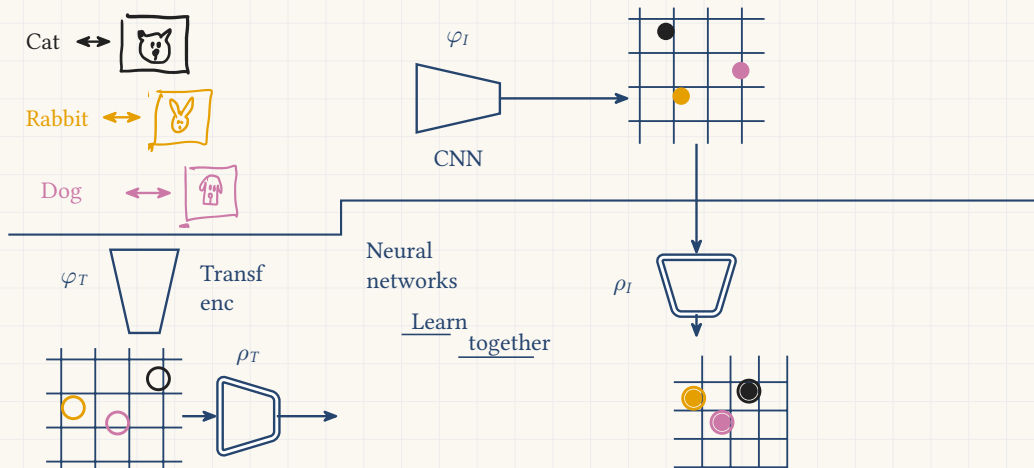
CLIP



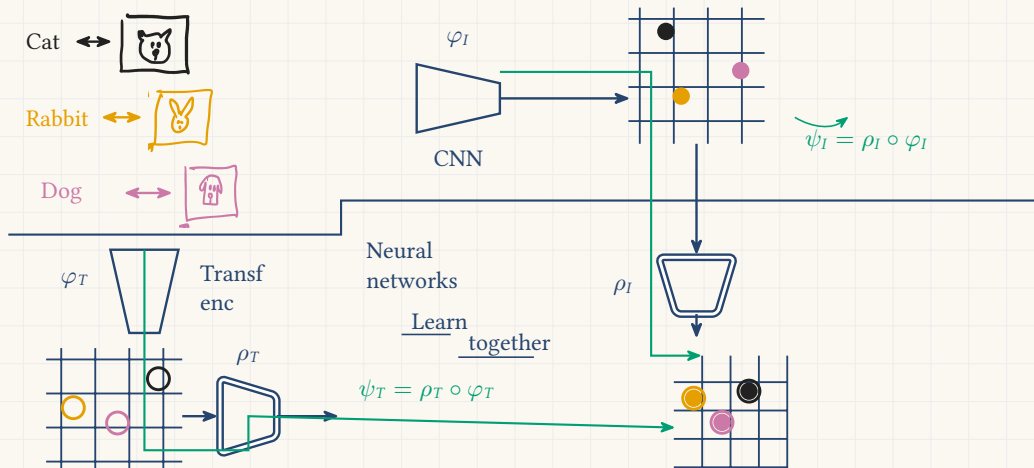
CLIP



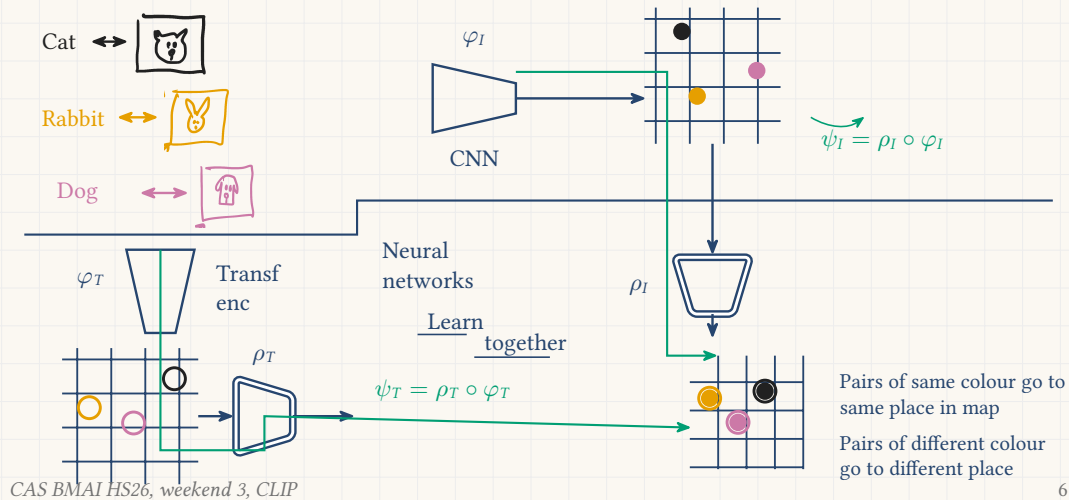
CLIP



CLIP



CLIP



2

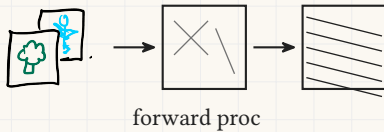
Section 2

Text-to-image translation with conditional diffusion models

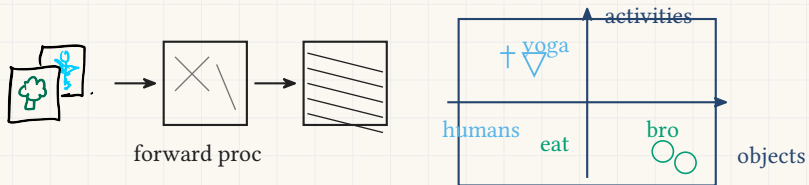
DALL-E



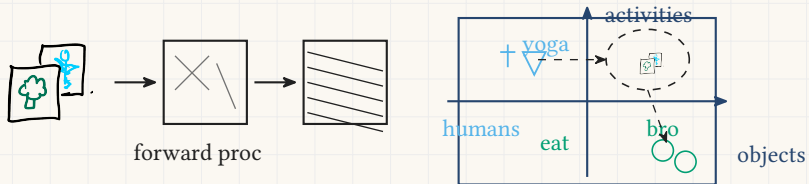
Fabricating images



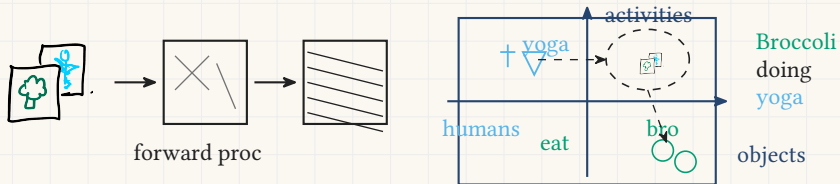
Fabricating images



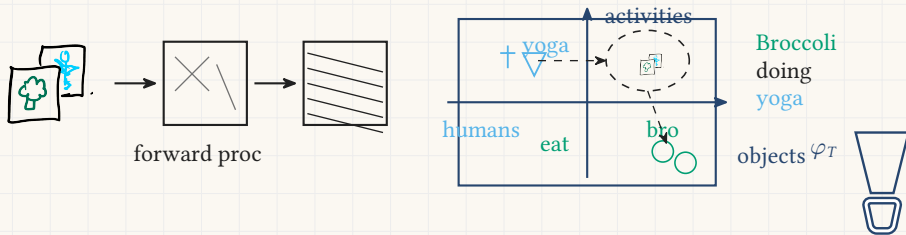
Fabricating images



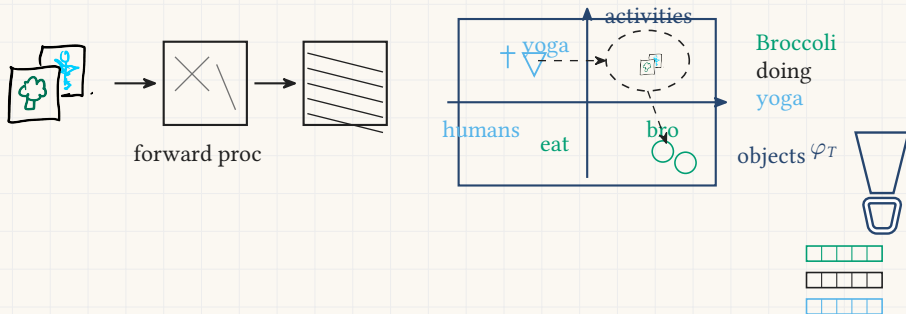
Fabricating images



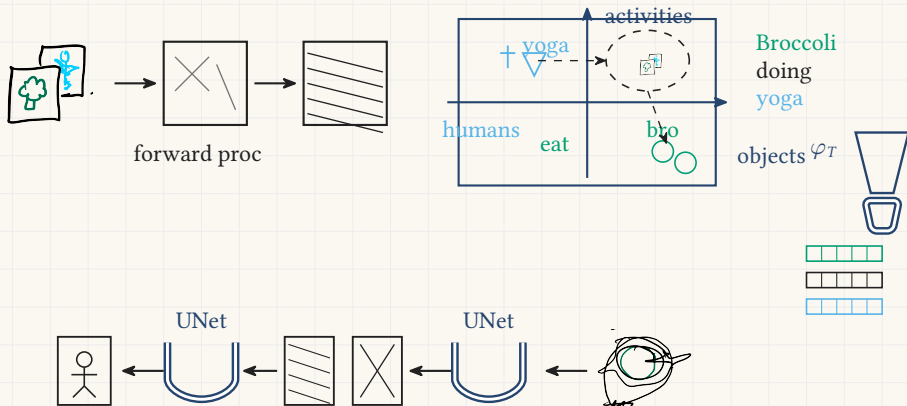
Fabricating images



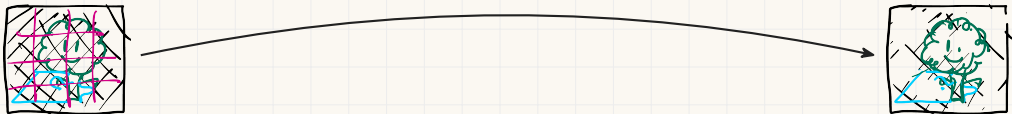
Fabricating images



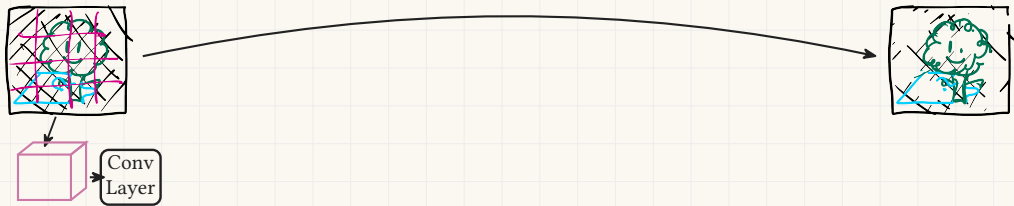
Fabricating images



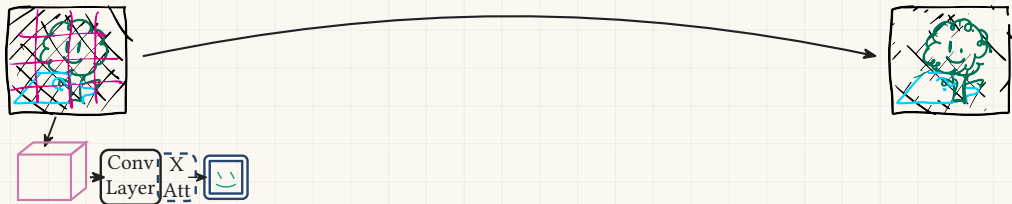
UNet, one step in reverse



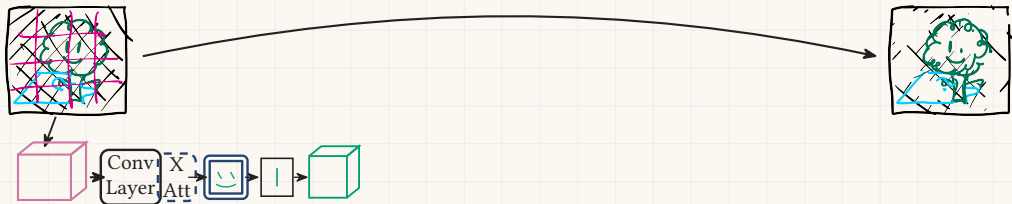
UNet, one step in reverse



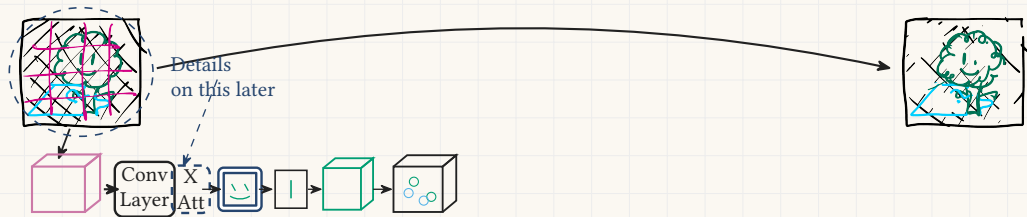
UNet, one step in reverse



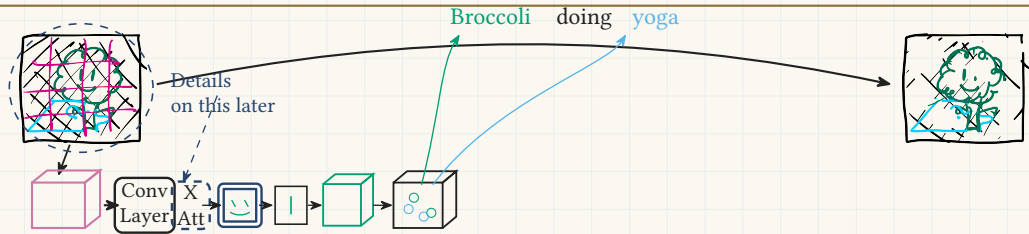
UNet, one step in reverse



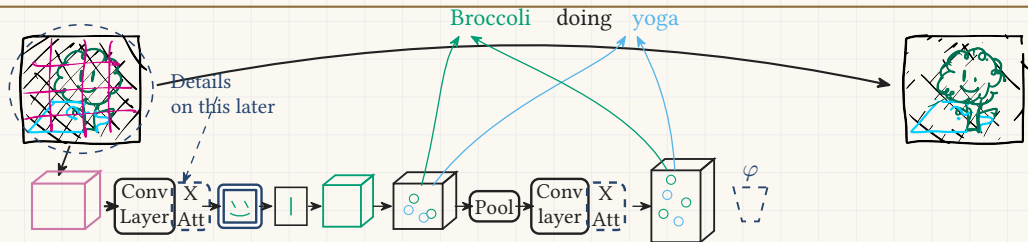
UNet, one step in reverse



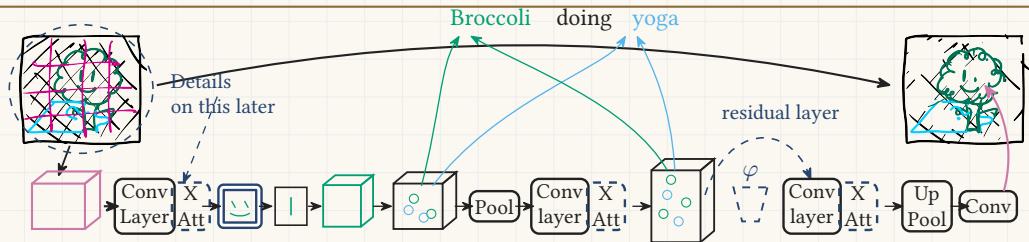
UNet, one step in reverse



UNet, one step in reverse



UNet, one step in reverse



How to train?

1) Data

$$\begin{array}{c} \{(x_1, t_1), \\ \dots \\ (x_n, t_n)\} \\ \uparrow \qquad \uparrow \\ \text{img} \quad \text{txt} \end{array}$$

How to train?

1) Data

$$\begin{array}{c} \{(x_1, t_1), \\ \dots \\ (x_n, t_n)\} \\ \uparrow \qquad \qquad \uparrow \\ \text{img} \qquad \text{txt} \end{array}$$

2) Models

How to train?

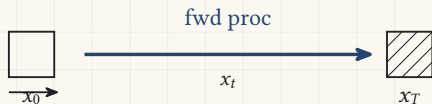
1) Data

$$\begin{array}{c} \{(x_1, t_1), \\ \dots \\ (x_n, t_n)\} \\ \uparrow \qquad \qquad \uparrow \\ \text{img} \qquad \text{txt} \end{array}$$

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

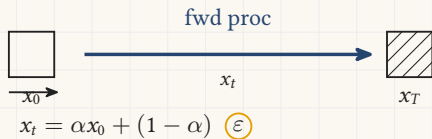
$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

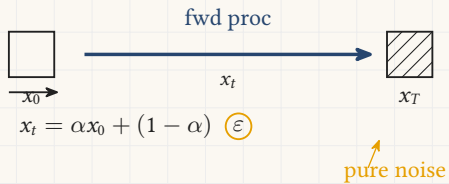
$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

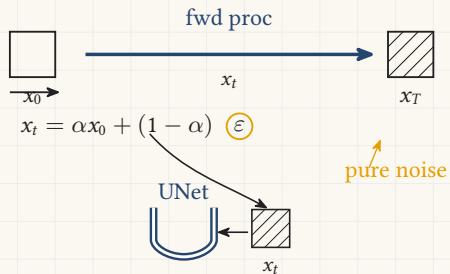
$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

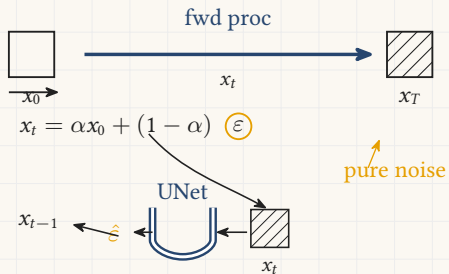
$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

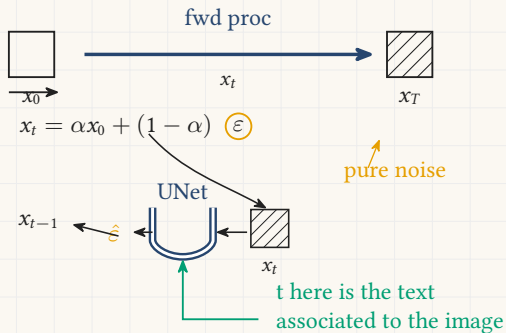
$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

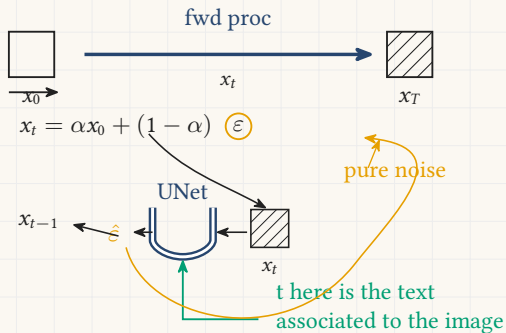
$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$

How to train?



1) Data

$$\{(x_1, t_1), \dots, (x_n, t_n)\}$$

img txt

2) Models

$$\mathcal{L}(\epsilon, \hat{\epsilon}) \approx \|\epsilon - \hat{\epsilon}\|^2$$